

E-Commerce Business Analysis using SQL – Project Report

Project Overview

This project focuses on analyzing e-commerce business data using SQL to generate meaningful business insights related to customers, products, and order behavior.

Technologies Used

- SQL
- MySQL / PostgreSQL
- Relational Database Management
- Business Analytics
- GitHub

Business Problem

E-commerce companies generate large volumes of transactional data but often struggle to extract actionable insights. This project helps analyze customer activity, product performance, and business growth using SQL queries.

Dataset Information

The project uses relational e-commerce datasets containing customer information, orders, products, and order-level transaction data.

Key Tables Used

- customers → Customer details
- orders → Order-level transactions
- order_items → Line-level purchase data
- products → Product and category details

Schema Relationship

Customers → Orders → Order_Items ← Products

SQL Concepts Used

- SELECT Statements
- WHERE Conditions
- GROUP BY & HAVING
- JOINS

- Aggregate Functions
- CASE WHEN
- Subqueries
- CTEs & Window Functions

Analysis Performed

The project analyzed customer activity trends, product category performance, revenue insights, and purchasing behavior to support business decision-making.

Business Insights

- Customer activity trends revealed business growth patterns.
- Certain product categories generated higher revenue.
- Product-level analysis supported inventory optimization.
- Customer behavior analysis supported retention planning.

Recommendations

- Improve customer retention strategies.
- Focus on high-performing product categories.
- Optimize inventory planning.
- Build automated reporting systems for regular KPI tracking.

Skills Demonstrated

- SQL Query Writing
- Relational Database Design
- Data Cleaning
- Business Analytics
- Insight Generation
- Problem Solving

Conclusion

This project demonstrates practical SQL analytics skills by transforming raw transactional data into meaningful business insights that support data-driven decision-making in e-commerce businesses.

GitHub Repository

<https://github.com/SaurabhVerma-art/ecommerce-business-analysis-SQL>